MIDTERM EXAM

Course Title	General Physics II	Course Code	30201102		
Semester	Second Semester	Academic Year	2025-2026		
Exam Date	18/4/2026	Day	saturday	Time	11 – 11:50

Student Name		Student ID No.		Section	

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide by the exam policy. **Student's signature:** _____

Number of Questions	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
Part One	Multiple Choice Questions	CLIO,1,2 and 7,8	9(30%)			
Problem 1	Essay	CLIO 1 and 2	3(10%)			
Problem 2	Essay	CLIO 3 and 4	4(13.3%)			
Problem 3	Essay	CLIO 5	3(10%)			
Problem 4	Essay	CLIO 6	4(13.3%)			
Problem 5	Essay	CLIO 7	4(13.3%)			
Problem 6	Essay	CLIO 8	3(10%)			
Total			30(100%)			------ /30

Physical Constants

Coulomb constant $k = \frac{1}{4\pi \in_o} = 8.987 \times 10^9\ N.\frac{m^2}{C^2}$

Permittivity of free space $\in_o = 8.854 \times 10^{-12}\ C^2/N.m^2$,

Electron mass $= 9.109 \times 10^{-31}\ kg$, Proton mass $= 1.672 \times 10^{-27}\ kg$,

Electron charge $= 1.602 \times 10^{-19}\ C$ The mathematical constant $\pi = 3.14$

Acceleration due to gravity $g = 9.80\ m/s^2$

الجزء الأول يتكون من (6) أسئلة إختيار من متعدد (9 Marks)

Q.1) A point charge Q is placed on the x-axis at the origin. An identical point charge is placed on the x-axis at x = –1.0 m and another at x = +3.0 m. If Q = 40 μC, what is the magnitude of the electrostatic force on the charge at x = +3.0 m?

A. 29.6 N **B. 2.5 N** C. 10.1 N D. 18.0 N E. 5.2 N

Q.2) Two particles, having oppositely signed charges of 8 nC, are placed at two of the vertices of an equilateral triangle (length of each side = 2.0 m). What is the magnitude of the electric field at the third vertex of the triangle?

A. 27 N/ C B. 36 N/ C C. 23 N/ C D. 12 N/ C **E. 18 N/ C**

Q.3) A particle (q = 3.0 mC, m = 20 g) has a speed of 20 m / s when it enters a region where the electric field has a constant magnitude of 80 N / C and a direction that is the same as the velocity of the particle. What is the speed of the particle 2.0 s after it enters this region?

A. 68 m/ s **B. 44 m/ s** C. 56 m/ s D. 80 m/ s E. 36 m/ s

Q.4) A uniform linear charge density of 4.0 nC / m is distributed along the entire x-axis. Consider a spherical (radius = 6.0 cm) surface centered on the origin. Determine the electric flux through this surface.

A. 68 $N.m^2/C$ B. 23 $N.m^2/C$ C. 45 $N.m^2/C$ D. 36 $N.m^2/C$ **E. 54 $N.m^2/C$**

Q.5 Points A [at (3, 6) m] and B [at (8, –3) m] are in a region where the electric field is uniform and given by E = 16i N / C. What is the electric potential difference $V_A - V_B$?

A. +60 V B. -60 V **C. +80 V** D. -80 V E. +50 V

Q.6 Three identical point charges (+20 μC) are placed at the corners of an equilateral triangle with sides of 5.0 m in length. What is the electric potential energy in the entire system?

A. 2.2J B. 1.2 J C. 3.6 J D. 4.8 J E. 5.4 J

الجزء الثاني يتكون من (6) أسئلة حسابية وحل السؤال مع الخطوات:

Problem.1) **A long nonconducting cylinder (radius = 12 cm) has a charge of uniform density (5.0 nC / m^3) distributed throughout its volume. Determine the magnitude of the electric field 8 cm from the axis of the cylinder. (3 Marks)**

$$E = \frac{\rho r}{2\epsilon_0} \quad (1)$$

$$= \frac{5\times10^{-9}\times 0.08}{2\times 8.85\times10^{-12}} \quad (1)$$

$$= 22.6 \ N/C \quad (1)$$

Problem.2) Over a certain region of space, the electric potential is $V = 5xyz - 3x^2y + 2yz^2$. Find the magnitude of the z-component of the electric field at the point (1, 1, -1)? (4 Marks)

$$E_z = -\frac{dV}{dz} = -\frac{d}{dz}[5xyz - 3x^2y + 2yz^2] \quad (1)$$

$$= -[5xy + 4yz] \quad (1)$$

$$E_z(1,1,-1) = -[5(1)(1) + 4(1)(-1)] \quad (1)$$

$$= -[5 - 4] = -1 \ N/C \quad (1)$$

Problem.3) A charge of 0.50 nC is placed at the center of a cube that measures 4.0 m along each edge. What is the electric flux through two faces of the cube? (3 Marks)

$$\Phi = \frac{Q_{in}}{\epsilon_0} \quad (1) = \frac{0.5\times10^{-9}}{8.85\times10^{-12}} = 56.50 \ N\cdot m^2/C \quad (1)$$

$$\Phi \text{ through 2-faces} = \frac{56.50}{3} = 18.83 \ N\cdot m^2/C \quad (1)$$

Problem.4) A proton is released from rest at the origin in a uniform electric field in the positive x direction with magnitude 750 N/C. What is the change in the electric potential energy of the proton–field system measured in electron volt when the proton travels to x = 4.50 m? (4 Marks)

$\Delta U = q\Delta V = -q\vec{E}\cdot\vec{S}$ (1)

$= -1.6\times10^{-19}(750)(4.5)\cos(0)$ (1)

$= -5.4\times10^{-16}$ J (1)

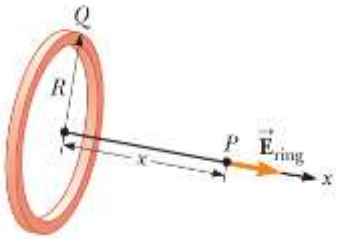
$= -3375$ e.V (1)

Problem.5) A uniformly charged ring of radius 10.0 cm has a total charge of 75.0 µC. Find the electric field on the axis of the ring at 20.0 cm. (3 Marks)

$E_{ring} = \dfrac{KQx}{(x^2+a^2)^{3/2}}$ (1)

$= \dfrac{9\times10^{9}\times75\times10^{-6}\times0.2}{(0.2^2+0.1^2)^{3/2}}$ (1)

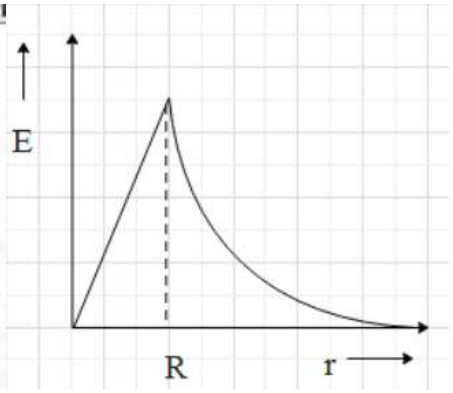
$\cong 12\times10^{6}$ N/C (1)



Problem.6) The following graph shows the relationship between the electric field and the distance of the point from the center of a charged sphere, where the field is measured. Determine whether the sphere is a conductor or an insulator, and explain your answer. (3 Marks)

insulator (1), E inside an insulator directly proportional to r and outside it inversely proportional to r^2. (2)

Q inside an insulator ≠ zero



Good Luck